Liveness Timeline for Client Request v

Step 1: Client→Switch $\mathsf{SwitchClientRequest}(v)$
Precondition: $v \notin \mathit{pendingRequests}[\mathit{Switch}]$
Postcondition: $v \in \mathit{pendingRequests}'[\mathit{Switch}]$
Temporal Property:

$$\Diamond\,(v \in \mathit{pendingRequests}[\mathit{Switch}])$$

Step 2: Switch→Servers $\mathsf{SwitchDisseminate}(v)$
Precondition: $v \in \mathit{pendingRequests}[\mathit{Switch}]$
Postcondition:

$$\forall s \in \mathit{Server} :\ v \in \mathit{pendingRequests}'[s]\ \wedge\ v \notin \mathit{pendingRequests}'[\mathit{Switch}]$$

Temporal Property:

$$(v \in \mathit{pendingRequests}[\mathit{Switch}])\ \leadsto\ \Diamond\,\forall s.\, v \in \mathit{pendingRequests}[s]$$

Step 3: Leader Election $\mathsf{BecomeLeader}(L)$
Precondition:

$$\mathit{votesGranted}[L] \in \mathit{Quorum}\ \wedge\ \mathit{leaderCount}[L] < MaxBecomeLeader$$

Postcondition:

$$\mathit{state}'[L] = \mathit{Leader}\ \wedge\ \mathit{currentTerm}'[L] = T$$

Temporal Property:

$$\Diamond\,\exists L.\,(\mathit{state}[L] = \mathit{Leader})$$

Step 4: Leader orders $\mathsf{LeaderOrderRequest}(L, v)$
Precondition:

$$\mathit{state}[L] = \mathit{Leader}\ \wedge\ v \in \mathit{pendingRequests}[L]$$

Postcondition:

$$\mathit{log}'[L] = \mathit{Append}\big(\mathit{log}[L], [\mathit{term}, \mathit{value} = v, \mathit{payload} = v]\big)\ \wedge\ v \notin \mathit{pendingRequests}'[L]$$

Temporal Property:

$$\big(\mathit{state}[L] = \mathit{Leader} \wedge v \in \mathit{pendingRequests}[L]\big)\ \leadsto\ \Diamond\,\exists \mathit{idx}.\ \mathit{log}[L][\mathit{idx}].\mathit{value} = v$$

Step 5: Send metadata $\mathsf{AppendEntries}(L, F)$
Precondition:

$$\mathit{nextIndex}[L][F] \leq \mathrm{Len}(\mathit{log}[L])$$

Postcondition:

$$\mathit{messages}' = \mathit{messages} \cup \{\, m \mid m.\mathit{mentries} = \langle\langle \mathit{term}, \mathit{value} = v\rangle\rangle\}$$

Temporal Property:

$$\big(\exists \mathit{idx}.\, \mathit{log}[L][\mathit{idx}].\mathit{value} = v\big)\ \leadsto\ \Diamond\,\exists m.\,(m \in \mathit{messages} \wedge m.\mathit{dest} = F \wedge m.\mathit{mentries}[1].\mathit{value} = v)$$

Step 6: **AE arrival** $\mathsf{HandleAppendEntriesRequest}(F, L, m)$
Precondition:

$$m \in messages \;\wedge\; m.mtype = AppendEntriesRequest \;\wedge\; m.mentries \neq \langle\rangle$$

Postcondition: branches 7 and 8 handle payload present/missing
Temporal Property:

$$\Diamond\,(m.mentries \neq \langle\rangle \wedge m \in messages) \;\leadsto\; \Diamond\,\mathsf{HandleAppendEntriesRequest}(F, L, m)$$

Step 7: **Reconstruct (present)**
$\mathsf{HandleAppendEntriesRequest}$ branch when payload present
Precondition:

$$v \in pendingRequests[F] \;\wedge\; m.mentries[1].value = v$$

Postcondition:

$$log'[F] = Append(log[F], fullEntry) \;\wedge\; v \notin pendingRequests'[F]$$

Temporal Property:

$$(v \in pendingRequests[F] \wedge AE\ delivered) \;\leadsto\; \Diamond\,\exists i.\, log[F][i].payload = v$$

Step 8: **RecoveryRequest (missing)**
$\mathsf{HandleAppendEntriesRequest}$ branch when payload missing
Precondition:

$$v \notin pendingRequests[F] \;\wedge\; m.mentries[1].value = v$$

Postcondition:

$$v \in missingRequests'[F] \;\wedge\; \mathsf{RecoveryRequest}(v) \in messages$$

Temporal Property:

$$(v \notin pendingRequests[F] \wedge AE\ delivered) \;\leadsto\; \Diamond\,(v \in missingRequests[F] \wedge RecoveryRequest\ sent)$$

Step 9: **Leader RecoveryResponse** $\mathsf{HandleRecoveryRequest}(L, F, m)$
Precondition:

$$m.mtype = RecoveryRequest \;\wedge\; v \in missingRequests[F] \;\wedge\; v \in log[L]$$

Postcondition:

$$\mathsf{RecoveryResponse}(v) \in messages$$

Temporal Property:

$$(v \in missingRequests[F]) \;\leadsto\; \Diamond\,\exists m.\,(m.mtype = RecoveryResponse \wedge m.mRequestValue = v)$$

Step 10: **Follower Recovery** $\mathsf{HandleRecoveryResponse}(F, L, m)$
Precondition:

$$m.mtype = RecoveryResponse \;\wedge\; m.mRequestValue = v$$

Postcondition:

$$v \in pendingRequests'[F] \;\wedge\; v \notin missingRequests'[F]$$

Temporal Property:

$$(\exists m.\, m.mtype = RecoveryResponse \wedge m.mRequestValue = v) \;\leadsto\; \Diamond\,(v \in pendingRequests[F])$$

Step 11: Commit $\mathsf{AdvanceCommitIndex}(L)$
Precondition:

$$\exists Q \subseteq Server: \ |Q| > \frac{|Server|}{2} \ \wedge \ \forall q \in Q. \, matchIndex[L][q] \geq idx_v \ \wedge \ log[L][idx_v].term$$

$= \text{currentTerm[L]}$

Postcondition:

$$commitIndex'[L] \geq idx_v$$

Temporal Property:

$$(quorum\_ack(idx_v)) \ \leadsto \ \Diamond \, (commitIndex[L] \geq idx_v)$$

Global Liveness Theorem

$$\boxed{\Diamond \, \exists \, i \in Server, \ idx: \ log_i[idx].value = v \ \wedge \ idx \leq commitIndex_i}$$

(Assuming weak fairness of enabled actions and eventual delivery of all messages.)